Dylan Laplace Mermoud

Assistant professor in algorithmic and algebraic game theory
Team 'Décision', LIP6, Sorbonne Université

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Areas of interest

Cooperative game theory, algebraic combinatorics, quantum algorithms, Coxeter groups.

Employment

Assistant professor Sept. 2026 – now
LIP6, Sorbonne Université

Postdoctoral researcher Oct. 2023 – Aug. 2026
Unité de Mathématiques Appliquées, ENSTA, Institut Polytechnique de Paris

▷ Supervisors: Andrea Simonetto and Sourour Elloumi

Teaching and research associate Sept. 2022 – Aug. 2023
Centre d'Économie de la Sorbonne, Université Paris I Panthéon-Sorbonne

Doctoral contract employee Sept. 2019 – Aug. 2022
Centre d'Économie de la Sorbonne, Université Paris I Panthéon-Sorbonne

Education

Doctoral degree in mathematics Sept. 2019 – Sept. 2023
Université Paris I Panthéon-Sorbonne

▷ Title: "Geometry of Set Functions in Game Theory: Combinatorial and Computational Aspects"
▷ Supervisors: Michel Grabisch and Peter Suhdölter

Master degree in mathematics Sept. 2017 – June 2019
Sorbonne Université

▷ Jointly accredited by École Polytechnique and École Nationale des Ponts et Chaussées.

Bachelor degree in mathematics Sept. 2014 – June 2017
Sorbonne Université

Publications, proceedings, and preprints

Publications

12. Sampled-based guided quantum walk: Non-variational quantum algorithm for combinatorial optimization
Nzongani, U., Laplace Mermoud, D., Di Molfetta, G., Simonetto, A.
Quantum, vol. 10, p. 2234 • 2025 • 27 pages • *arXiv.2509.15138v3*

11. The algebraic structures of social organizations: the operad of cooperative games
Laplace Mermoud, D., Roca i Lucio, V.
Algebraic Combinatorics • 2026 • 49 pages • *arXiv.2507.01969v2*

10. Variational quantum algorithms for permutation-based combinatorial problems: Optimal ansatz generation with applications to quadratic assignment problems and beyond
Laplace Mermoud, D., Simonetto, A., Elloumi, S.
Quantum, vol. 10, p. 1998 • 2026 • 36 pages • *arXiv.2509.05981v3*

9. Attraction of the core and the cohesion flow
Laplace Mermoud, D.
Theory and Decision, vol. 99, pp. 377–392 • 2025 • 15 pages • *arXiv.2507.02918*

8. Minimal balanced collections and their application to core stability and other topics of game theory
Laplace Mermoud, D., Grabisch, M., Sudhölter, P.
Discrete Applied Mathematics, vol. 341, pp. 60–81 • 2023 • 32 pages • *arXiv.2507.05898*

Proceedings

7. Scaling QAOA: transferring optimal adiabatic schedules from small-scale to large-scale variational circuits
Nzongani, U., Laplace Mermoud, D., Braida, A.
2026 IEEE QCCL • 2026 • 6 pages • *arXiv.2602.14986*

6. Combinatorics on social configurations
Laplace Mermoud, D., Popoli, P.
Electronic Proceedings in Theoretical Computer Science, vol. 403, pp. 128–133 • 2024 • 6 pages

Preprints

5. Circuit Design Problems in Binary Polynomial Quantum Optimization
Laplace Mermoud, D., Watel, D., Elloumi, S., Simonetto, A.
arXiv preprint arXiv.2610.13283 • 2026 • 26 pages

4. On the closest balanced game
Garcia-Segador, P., Grabisch, M., Laplace Mermoud, D., Miranda, P.
arXiv preprint arXiv.2601.15318 • 2025 • 31 pages

3. Diffusion in dynamic networks with time-varying inputs to allocate responsibility
van den Ende, R., Laplace Mermoud, D.
arXiv preprint arXiv.2301.02728 • 2025 • 25 pages

2. Projection onto the core: An optimal reallocation to correct market failure
Laplace Mermoud, D.
arXiv preprint arXiv.2411.11810 • 2024 • 30 pages

1. On the formation of steady coalitions
Laplace Mermoud, D.
arXiv preprint arXiv.2410.05087 • 2024 • 21 pages

Research visits

- ▷ Michel Grabisch at the Algorithmic game theory in socioeconomics (AGATE) center
Charles University, Prague, Czech Republic • one week • 2026 (forthcoming)
- ▷ Māris Ozols and Krystal Guo at the Discrete Mathematics and Quantum Information group
Universiteit van Amsterdam and CWI, The Netherlands • two weeks • 2025
- ▷ Stefan Wörner at the Quantum Computational Science group
IBM Research Europe - Zurich, Switzerland • one week • 2025
- ▷ Victor Roca i Lucio at the Laboratory for Topology and Neurosciences
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland • one week • 2023
- ▷ Pierre Popoli at the Laboratory of Discrete Mathematics
Université de Liège, Belgium • one week • 2023

Teaching

Lectures

- ▷ Quantum computing — 2025 – 2026
  MPRO & ensIIE • Master 2 in informatics • 6h
- ▷ Game theory — 2024 – 2026
  ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 7h30 • Lecture notes

Tutorials

- ▷ Probabilistic and statistical models and algorithms — 2026 – now
  Sorbonne Université • Master 1 in informatics • 40h
- ▷ Modelling, optimization, graphs, and linear programming — 2026 – now
  Sorbonne Université • Master 1 in informatics • 20h
- ▷ Algorithms II — 2026 – now
  Sorbonne Université • Bachelor 3 in informatics • 40h
- ▷ Data structures and C langage — 2026 – now
  Sorbonne Université • Bachelor 2 in informatics • 20h
- ▷ Quantum computing — 2025 – 2026
  MPRO & ensIIE • Master 2 in informatics • 15h
- ▷ Game theory — 2024 – 2026
  ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 10h
- ▷ Graph theory — 2023 – 2026
  ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 16h30
- ▷ Linear programming — 2020 – 2023
  Université Paris I Panthéon-Sorbonne • Master 1 in mathematics • 24h
- ▷ Topology and differential calculus — 2022 – 2023
  Université Paris I Panthéon-Sorbonne • Bachelor 3 in mathematics/informatics • 42h
- ▷ Microeconomics — 2022 – 2023
  Université Paris I Panthéon-Sorbonne • Bachelor 3 in mathematics/informatics • 48h
- ▷ Linear algebra — 2022 – 2023
  Université Paris I Panthéon-Sorbonne • Bachelor 2 in mathematics/informatics • 30h
- ▷ Linear algebra — 2019 – 2023
  Université Paris I Panthéon-Sorbonne • Bachelor 3 in economics • 36h
- ▷ Python programming — 2019 – 2020
  Université Paris I Panthéon-Sorbonne • Bachelor 1 in mathematics/informatics • 48h

Supervision

- ▷ Paul-Henri Djoko (2026), Master 1 student, on reinforcement learning in game theory.
- ▷ Bruno Galibert (2025), Master 1 student, on quantum circuit simplification.
- ▷ Qui Loc Phan (2024), Master 1 student, on quantum circuit design and combinatorial optimization.

Awards and fundings

- Laureate of the *Quantum-Saclay* grant, one internship funding (2025).
- Laureate of the *Passeport pour le Master* scholarship from Sorbonne Université Foundation, fully funded 5 years of study (2014-2019).

Responsibilities

- Member of the board of directors of Sorbonne Université (2018-2020), member of the board of the mathematics department of Sorbonne Université (2018-2020).

Refereeing

- Journal of Mathematical Economics, 4OR - A Quarterly Journal of Operations Research, Annals of Operations Research (x2), IEEE International Conference on Quantum Control, Computing and Learning (x2), CEMRACS 2025 – Quantum Computing, Operations Research Letters (x3), Fuzzy Sets and Systems (x2), International Journal of Game Theory.

Communications

- Session organization
  1. "Quantum computing for combinatorial optimization" at Programme Gaspard Monge (PGMO) days 2025, Palaiseau, France (2025).
- Upon invitation
  2. SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom (2026).
  3. 18th Ordered Structures in Games and Decisions in Michel Grabisch's honor, Paris, France (2025).
  4. 16th Ordered Structures in Games and Decisions conference, Paris, France (2023).
  5. 22nd Society for the Advancement of Economic Theory (SAET) conference, Paris, France (2023).
  6. GO22: Games and Optimization 2022, Saint-Étienne, France (2022).
- Conferences
  7. 27th ROADEF congress, Tours, France (2026).
  8. 20th European Meeting on Game Theory (SING20), Maastricht, The Netherlands (2025).
  9. International Conference in Quantum Computing (ICoQC2025), Paris, France (2025).
  10. 26th ROADEF congress, Champs-sur-Marne, France (2025).
  11. Programme Gaspard Monge (PGMO) days 2024, Palaiseau, France (2024).
  12. 19th European Meeting on Game Theory (SING19), Besançon, France (2024).
  13. 13th Random Generation of Combinatorial Structures. Polyominoes and Tiling (GASCom), Bordeaux, France (2024).
  14. 18th European Meeting in Game Theory (SING18), Messina, Italy (2023).
  15. 26th Coalition Theory Network (CTN) conference, Bielefeld, Germany (2023).
  16. Programme Gaspard Monge (PGMO) days 2021, Palaiseau, France (2021).
  17. 16th European Meeting in Game Theory (SING16), Granada (remote), Spain (2021).
- Seminars
  18. Universiteit van Amsterdam, Discrete Mathematics and Quantum Information group seminar, Amsterdam, The Netherlands (2025).

19. IBM Research Europe - Zurich, Quantum Optimization project, Zurich, Switzerland (2025).
20. Université Aix-Marseille, CANA research group seminar, Marseille, France (2025).
21. C12, R&D group seminar, Paris, France (2025).
22. Université Paris Cité, IRIF quantum team seminar, Paris, France (2025).
23. École Polytechnique, PhIQus team seminar, Palaiseau, France (2025).
24. Télécom Paris, Quriosity team seminar, Palaiseau, France (2025).
25. Quandela, R&D group seminar, Massy, France (2025).
26. Conservatoire National des Arts et Métiers, Combinatorial Optimization (OC) group seminar, Paris, France (2025).
27. ENSTA - Institut Polytechnique de Paris, Optimization and Command (OC) group seminar, Palaiseau, France (2025).
28. Centre d'Économie de la Sorbonne, "Mathématique Discrète, Optimisation, Décision" seminar, Paris, France (2024).
29. Université de Franche-Comté, "young researchers" day, Besançon, France (2024).
30. Institut Henri Poincaré, Junior Parisian seminar in game theory, Paris, France (2023).
31. ENSTA - Institut Polytechnique de Paris, Optimization and Command (OC) group seminar, Palaiseau, France (2023).
32. Paris School of Economics, Doctorissimes, Paris, France (2022).
33. Centre d'Économie de la Sorbonne, Maths department days, Paris, France (2022).
34. Centre d'Économie de la Sorbonne, "Network and Games" seminar, Paris, France (2022).
35. Centre d'Économie de la Sorbonne, PhD seminar, Paris, France (2021).
36. Institut Élie Cartan, PhD seminar, Nancy, France (2021).